

Homework — 2.2.2 Computational Methods

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 2.2.2 lesson. Show traces/working.

Part A — This week's topic (~14 marks)

1. State **two** characteristics a problem must have for it to be **solvable computationally**. [2] (AO1)

2. Explain the difference between **decomposition** and **divide and conquer**. [2] (AO1)

3. A weather service represents temperature readings across a country as a colour-coded **heat map** rather than a long table of numbers.

(a) Name the computational method being used. [1] (AO1)

(b) Give **one** benefit of this method when analysing large data sets. [1] (AO1)

4. A delivery firm uses a **heuristic** to plan routes for its fleet rather than an exact method.

(a) State what is meant by a **heuristic**. [1] (AO1)

(b) Explain **why** a heuristic, rather than an exact algorithm, is appropriate for a problem such as the Travelling Salesman Problem. [2] (AO2)

5. Explain how **pipelining** can increase the throughput of a process, using an example. [2] (AO1)

6. Explain the difference between **representational abstraction** and **abstraction by generalisation**, giving **one** example of each. [3] (AO1)

Part B — Retrieval: keep it fresh (~6 marks)

7. (2.2.1) State the defining difference between a **function** and a **procedure**. [1]

8. (2.2.1) A recursive subroutine has no reachable **base case**. State the run-time error that results and explain why it occurs. [2]

9. (2.1) State what is meant by **thinking ahead**, and give **two** examples of where it is used when designing a system. **[3]**

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